Active Inference for Joint Port Selection and Pilot Allocation in Fluid Antenna Systems

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***Abstract*—*[DRAFT — rewrite last, once the results section is frozen.]* Fluid antenna systems (FAS) obtain spatial degrees of freedom by activating a small subset of a dense grid of candidate ports, but selecting that subset presupposes channel knowledge across ports that are never observed. Existing port-selection methods assume full or near-full channel state information (CSI) and treat pilot overhead as exogenous. We instead formulate port selection and pilot allocation jointly as *active inference*: a spatio-temporal generative model maintains a Gaussian belief over all N port channels, and the transmitter selects both the ports it serves and the ports it pilots by minimizing an expected free energy (EFE) that trades achievable rate against information gain and port-switching cost. [TODO: one sentence on the temporal model / online Doppler learning] On a 21×21 port grid, the proposed scheme attains a higher sum rate while piloting only 60% of the active ports than a memoryless baseline piloting all of them, and recovers 83% of the oracle rate when the Doppler rate is unknown and learned online.**

***Index Terms*—Fluid antenna systems, port selection, active inference, expected free energy, pilot allocation, partial CSI.**

I. Introduction

[TODO: Para 1: FAS context — ports, spatial DoF, RF-chain budget.]

[TODO: Para 2: the acquisition problem — scoring N ports needs CSI at N ports; overhead grows with the grid.]

[TODO: Para 3: prior work and the gap — learning-based selection (DRL, bandits) assumes full CSI; FAS channel estimation solves *estimation*, not the *decision*. No prior work allocates pilots as part of the selection decision.]

[TODO: Para 4: contributions, as a bulleted list.]

II. System Model

[TODO: Geometry and port grid; spatial correlation $\mathbf{R}$; multiuser downlink; hybrid transmit front end; per-slot protocol timeline.]

III. Generative Model and Belief

[TODO: Spatio-temporal state-space model; AR(p) temporal prior; Kalman recursion; partial observation via selection matrix; reduced-rank implementation.]

IV. Expected Free Energy and Selection

[TODO: EFE objective; pragmatic / epistemic / switching terms; submodularity and the greedy guarantee; joint selection of served and piloted ports.]

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V. Online Doppler Learning

[TODO: Matched-window ACF estimator; order selection; why a parametric Jakes fit fails in closed loop.]

VI. Numerical Results

[TODO: Setup table; pilot–rate trade-off figure; Doppler-learning figure; baselines.]

VII. Conclusion

[TODO:]

References

[1] K.-K. Wong, K.-F. Tong, Y. Zhang, and W. Zhongbin, "Fluid antenna system: A new spatial diversity paradigm," *IEEE Trans. Wireless Commun.*, 2021.
[2] "Switching-cost-aware deep reinforcement learning for dynamic port selection in fluid antenna systems," *IEEE Commun. Lett.*, 2024.
[3] "Online learning-induced port selection for fluid antenna in dynamic channel environment," 2024.
[4] "Successive bayesian reconstructor for channel estimation in fluid antenna systems," *IEEE Trans. Wireless Commun.*, 2025.
[5] "Fluid antenna with linear MMSE channel estimation for large-scale cellular networks," *IEEE Trans. Commun.*, 2023.
[6] "Channel estimation and reconstruction in fluid antenna system: Oversampling is essential," 2024.
[7] F. Sohrabi and W. Yu, "Hybrid digital and analog beamforming design for large-scale MIMO systems," *IEEE J. Sel. Topics Signal Process.*, 2016.
[8] X. Zhang, A. F. Molisch, and S.-Y. Kung, "Variable-phase-shift-based RF-baseband codesign for MIMO antenna selection," *IEEE Trans. Signal Process.*, 2005.
[9] K. Friston, F. Rigoli, D. Ognibene, C. Mathys, T. Fitzgerald, and G. Pezzulo, "Active inference and epistemic value," *Cognitive Neuroscience*, 2015.
[10] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher, "An analysis of approximations for maximizing submodular set functions—I," *Mathematical Programming*, vol. 14, no. 1, pp. 265–294, 1978.
[11] W. C. Jakes, *Microwave Mobile Communications*. Wiley, 1974.